Status	Finished
Started	Thursday, 27 August 2026, 2:28 PM
Completed	Thursday, 27 August 2026, 2:34 PM
Duration	6 mins 15 secs
Marks	9.00/11.00
Grade	**13.09** out of 16.00 (**81.82**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which medium is most suitable for **digital transmission** according to the lecture?

- a. A low-pass medium
- b. A carrier-modulated wireless channel only
- c. A band-pass medium only
- d. A medium that passes only one frequency
- e. A medium that blocks all low frequencies

The correct answer is: A low-pass medium

Question 2

Complete

Mark 1.00 out of 1.00

Which condition is **most likely to cause baseline wandering in a digital signal**?

- a. Using a line coding scheme with frequent transitions, such as Manchester encoding
- b. Using more voltage levels to represent data
- c. Frequent transitions between positive and negative voltage levels
- d. A long sequence of consecutive 0s or consecutive 1s
- e. Decreasing the duration of each bit interval

This wording matches the lecture more closely, which states that a long string of 0s or 1s can cause the baseline to drift and make decoding more difficult.

The correct answer is: A long sequence of consecutive 0s or consecutive 1s

Question 3

Complete

Mark 0.00 out of 1.00

Why is a **DC component undesirable** in digital transmission?

- a. It automatically causes the receiver clock to become faster than the sender clock.
- b. It converts the digital signal into an analog signal.
- c. It may be blocked by transmission systems that do not pass very low frequencies, causing signal distortion.
- d. It prevents the receiver from detecting any signal transitions.
- e. It increases the number of bits transmitted per second.

The correct answer is: It may be blocked by transmission systems that do not pass very low frequencies, causing signal distortion.

Question 4

Complete

Mark 1.00 out of 1.00

Which feature is essential for a digital signal to be **self-synchronizing**?

- a. It uses positive voltage levels to represent transmitted data.
- b. It uses an analog carrier to represent the transmitted bits.
- c. It contains a DC component during the transmission period.
- d. It maintains the same voltage level for several bit intervals.
- e. It includes timing information through transitions in the signal.

The correct answer is: It includes timing information through transitions in the signal.

Question 5

Complete

Mark 1.00 out of 1.00

Which combination best describes a good digital transmission scheme?

- a. Baseline wandering, DC component, synchronization
- b. Only positive voltage levels
- c. No synchronization but very high amplitude
- d. No baseline wandering, no DC component, and self-synchronization
- e. High DC component, no transitions, low bandwidth

The correct answer is: No baseline wandering, no DC component, and self-synchronization

Question 6

Complete

Mark 1.00 out of 1.00

What is a major problem associated with **unipolar encoding**?

- a. It can have a nonzero average amplitude, producing a DC component.
- b. It uses only negative voltage levels to represent transmitted bits.
- c. It requires two signal transitions to represent every transmitted bit.
- d. It requires four different voltage levels to represent binary data.
- e. It cannot use a voltage level to represent binary bit 1.

The correct answer is: It can have a nonzero average amplitude, producing a DC component.

Question 7

Complete

Mark 1.00 out of 1.00

Which statement correctly compares the **synchronization problems of NRZ-L and NRZ-I**?

- a. NRZ-L avoids synchronization problems completely, while NRZ-I does not provide any synchronization.
- b. Both NRZ-L and NRZ-I return to zero during each bit interval to maintain synchronization.
- c. Both NRZ-L and NRZ-I always provide transitions for every transmitted bit.
- d. NRZ-L may lose synchronization with long runs of 0s or 1s, while NRZ-I mainly has problems with long runs of 0s.
- e. NRZ-L mainly has problems with long runs of 0s, while NRZ-I mainly has problems with long runs of 1s.

The correct answer is: NRZ-L may lose synchronization with long runs of 0s or 1s, while NRZ-I mainly has problems with long runs of 0s.

Question 8

Complete

Mark 1.00 out of 1.00

A receiver clock is **0.2% faster** than the sender clock.

If the sender transmits at **500 kbps**, the receiver detects [1000] extra bits per second.

500000×0.002=1000

Question 9

Complete

Mark 2.00 out of 2.00

An NRZ-I system transmits data at **16 Mbps**. Assume c=0.5 and r=1.

The average signal rate is 8 MBaud and the minimum bandwidth 8 MHz

S = c N /r = 0.5*16/1 MBaud

Question 10

Complete

Mark 0.00 out of 1.00

A channel has a bandwidth of **4 MHz**. Assume c=0.5 and each symbol represents **2 bits** (r=2).

The maximum data rate is 12 Mbps

S = c * N / r
Nmax = 1/c *S r = 1/0.5 x 4M x 2 = 16 Mbps*